# IoT Smart Energy Metering and Real-Time Building Telemetry Report

**Document ID:** DOC_IOT_008 | **Type:** TECHNICAL_REPORT | **Publication:** 2024-10-05 | **Institution:** VESIT (VESIT Green Campus, Chembur)

## IOT SMART ENERGY METERING AND REAL-TIME BUILDING TELEMETRY REPORT

Document Reference: VESIT/TECH/IOT-ENERGY/2024/01

Project Reference: INIT_IOT_ENERGY_MGT (Campus IoT Energy Management)

Executing Departments: Department of Computer Engineering (DEP_CMPN) and Department of Artificial Intelligence & Data Science (DEP_AIDS).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

## 1. ARCHITECTURAL TOPOLOGY AND HARDWARE DEPLOYMENT

In March 2024, Vivekananda Education Society's Institute of Technology (VESIT Green Campus) deployed project INIT_IOT_ENERGY_MGT, a campus-wide digital telemetry network designed to provide granular, continuous monitoring of electrical energy flows. The infrastructure covers 32,500 m2 of institutional floor space across seven buildings: Academic Block A (BLD_ACA), Academic Block B (BLD_ACB), Computer Laboratories (BLD_LAB), Library (BLD_LIB), Auditorium (BLD_AUD), Canteen (BLD_CAN), and Administration Block (BLD_ADM).

The deployment comprises 45 industrial multi-function digital energy meters (Schneider Electric EM6400NG series) communicating over RS-485 serial buses using the open Modbus-RTU protocol. Meters are integrated across all Main Distribution Boards (MDBs) and Sub-Distribution Boards (SDBs). Five edge gateway nodes (Raspberry Pi 4 Compute Modules running industrial Linux) query the meter network every 15 seconds, aggregating active power (kW), reactive power (kVAR), line voltage (V), phase current (A), total harmonic distortion (THD), and power factor (PF). Normalized telemetry is packaged into lightweight JSON payloads and published via MQTT over the campus fiber-optic backbone to a secure local time-series server running InfluxDB and Grafana, mirrored directly into the EcoClock AI knowledge engine.

## 2. TELEMETRY INSIGHTS AND ANOMALOUS CONSUMPTION DISCOVERY

Analysis of over 18 million continuous telemetry records collected between April 1 and September 30, 2024, uncovered three substantial operational inefficiencies:

2.1 Computer Laboratories Standby Baselines (BLD_LAB):

Telemetry established that non-operational nighttime electrical draw (between 22:00 and 06:00) averaged 24.2 kW. Despite labs being unoccupied, over 380 desktop PC workstation power supplies remained plugged in quiescent sleep mode, drawing 18-22 W each. Concurrently, precision air conditioning units in server rooms were operating at sub-optimal setpoints of 19.0°C instead of the recommended 24.0°C.

2.2 Auditorium Parasitic Loads (BLD_AUD):

In the Auditorium, stage lighting step-down transformers, audio power amplifiers, and projection cooling fans drew 3.8 kW continuous parasitic power even when the hall was dark and locked between weekly events.

2.3 Academic Concurrency Demand Spikes (BLD_ACA and BLD_ACB):

Campus peak demand reached 410 kW between 11:30 and 14:00 on weekdays, caused by simultaneous operation of heavy lab computing, classroom split ACs, and Canteen kitchen ventilation.

## 3. AUTOMATED CONTROL INTERVENTIONS AND QUANTIFIED ENERGY CONSERVATION

To eliminate these identified baseload drains, the engineering team executed three firmware and control optimizations:

- Smart Contactor Bench Relays: Installed 16 automated Modbus-controlled relay banks across BLD_LAB. Timed firmware scripts automatically disconnect electrical supply to student workstation benches at 20:30 on weekdays and all weekend hours, eliminating 16.5 kW of phantom draw.

- Server Room Climate Optimization: Reprogrammed precision CRAC thermostats in the AI Research server room from 19.0°C to 24.0°C, delivering an immediate 14.2% reduction in compressor energy demand without compromising hardware thermal margins.

- Occupancy Automation in BLD_AUD: Integrated passive infrared (PIR) occupancy sensors to de-energize stage lighting transformers when the hall is vacant for over 15 minutes.

## 4. ENERGY AND CARBON REDUCTION IMPACT

Post-optimization telemetry over a verified 90-day period demonstrated a sustained 4.80% reduction in campus-wide monthly electricity consumption:

- Annualized Electricity Conservation: 98,000 kWh saved across campus facilities.

- Avoided Grid Emissions: Applying Central Electricity Authority factor EF_ELEC_GRID (0.82 kg CO2e/kWh), the IoT project directly abated 80,360 kg CO2e in 2024.

- Financial Return: Annual utility bill reductions equal INR 10,78,000 against a total IoT implementation expenditure of INR 8,40,000, establishing a simple payback period of only 9.3 months.

The telemetry infrastructure directly underpins target TRG_NET_ZERO and advances United Nations SDG 7, SDG 9, and SDG 12.

## INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_IOT_008**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*